

**Programme: B E – Computer Science and Engineering (AI&ML) &
Computer Science and Engineering (Cyber Security)
Internal Assessment – I**

TERM : CIE-1	COURSE NAME: COA
DATE :30-11-2024 TIME: 9:30 to 10:30AM	COURSE CODE : CI34/CY34
MAX MARKS: 30	PORTIONS : L1-L18 (UNIT 1 and UNIT 2)



Mobile Phones are banned

Instructions to Candidates: Answer any two full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Prove the identity $x+wx^1 = x+ w$ using truth table.	L3	CO1	8
B	Describe the working of clocked SR-flip flop with neat circuit diagram and truth table.	L2	CO1	7
2.a	Simplify the following four variable expression $f(A,B,C,D) = \sum m(3,5,7,8,10,11,12,13)$ using K-map/Boolean algebra.	L3	CO2	5
B	Explain CMOS realization of a NAND gate with circuit diagram and truth table.	L2	CO1	5
C	With a neat diagram explain the data transfer between memory and processor?	L2	CO2	5
3.a	Construct the circuit x-NOR gate using standard basic gates?	L3	CO1	5
B	Write a program that can evaluate the expression $TOTAL = A*B+C+D+E*F$. Assume 1-address mode processor.	L3	CO2	6
C	With an example describe the concepts of Big-endian and Little-endian address assignments?	L2	CO2	4

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create